SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

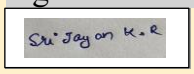
ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	**CSA0314**	**Course Title**	**Data Structures**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Concept Mapping**
Weightage	**35% of CIA**	**Total Marks**	**100**
CO4 Statement	***Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.* (BL2, BL3)**		
Student Name	**K.R. Sri Jayan**	**Reg. No.**	**1921260271**
Section/Batch	**Slot C /**	**Date**	**03-10-2026**

Code of Conduct:

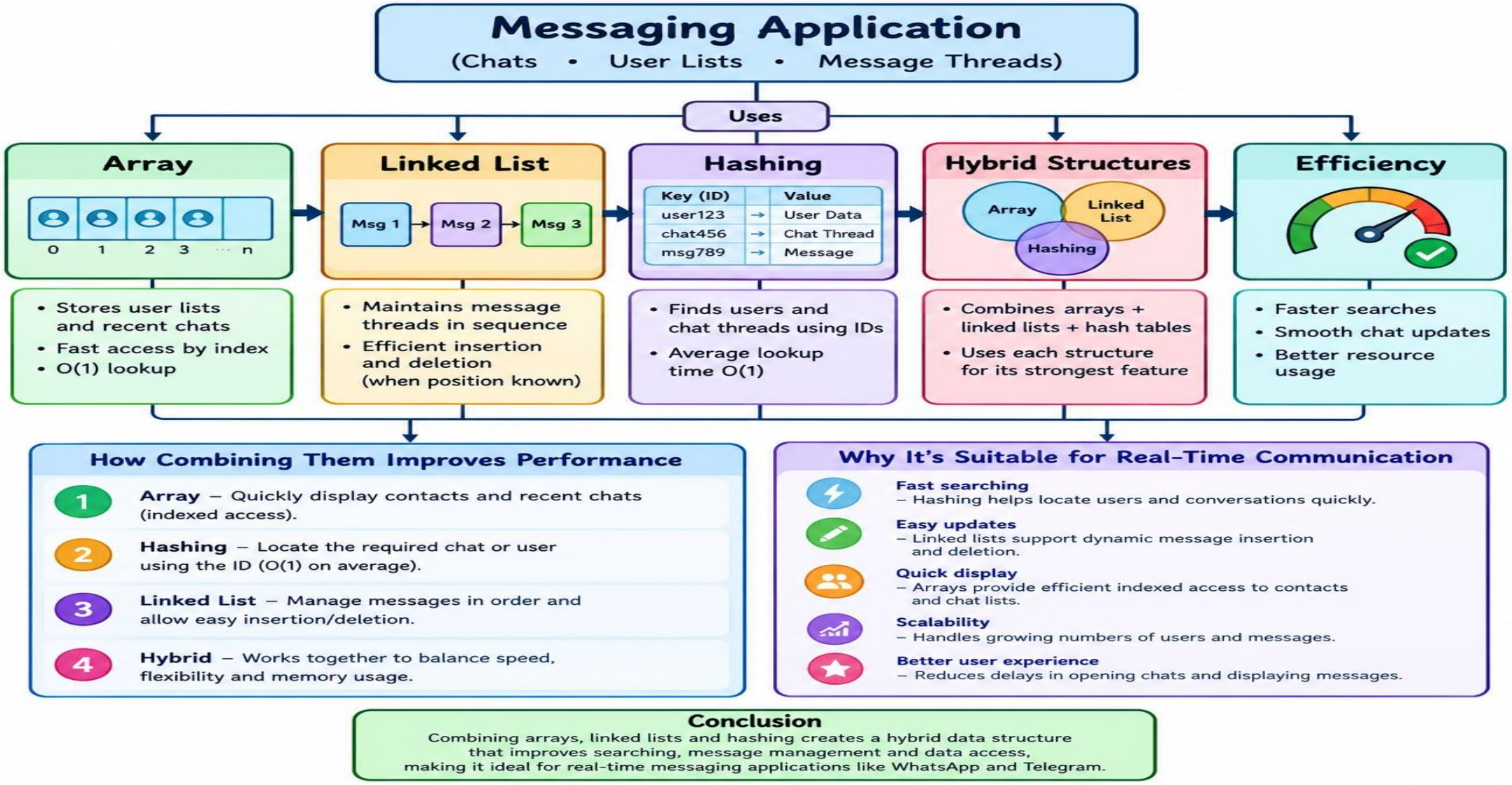
I,__Sri Jayan K.R_____, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

CONCEPT MAP


Messaging Application
(Chats • User Lists • Message Threads)
Uses
Array
0 1 2 3 ··· n
Linked List
Msg 1
Msg 2
Msg 3
Hashing
Key (ID) Value
user123 → User Data
chat456 → Chat Thread
msg789 → Message
Hybrid Structures
Array
Linked List
Hashing
Efficiency
• Stores user lists and recent chats
• Fast access by index
• O(1) lookup
• Maintains message threads in sequence
• Efficient insertion and deletion (when position known)
• Finds users and chat threads using IDs
• Average lookup time O(1)
• Combines arrays + linked lists + hash tables
• Uses each structure for its strongest feature
• Faster searches
• Smooth chat updates
• Better resource usage
How Combining Them Improves Performance
1 Array – Quickly display contacts and recent chats (indexed access).
2 Hashing – Locate the required chat or user using the ID (O(1) on average).
3 Linked List – Manage messages in order and allow easy insertion/deletion.
4 Hybrid – Works together to balance speed, flexibility and memory usage.
Why It's Suitable for Real-Time Communication
Fast searching – Hashing helps locate users and conversations quickly.
Easy updates – Linked lists support dynamic message insertion and deletion.
Quick display – Arrays provide efficient indexed access to contacts and chat lists.
Scalability – Handles growing numbers of users and messages.
Better user experience – Reduces delays in opening chats and displaying messages.
Conclusion
Combining arrays, linked lists and hashing creates a hybrid data structure that improves searching, message management and data access, making it ideal for real-time messaging applications like WhatsApp and Telegram.